

Route G-R6: Separating Thermal Response from a Gravitational Clock Baseline

A fixed-input two-temperature, two-potential control

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Abstract

Retain G-R4's E2/E3 thermal action and inputs and G-R5's external E3 reference. Add an independently calibrated gravitational-potential contrast. A four-setting logarithmic comparison separates thermal, gravitational and interaction responses without fitting a time law. Matched local radiation states cancel even uncertain fixed thermal shifts in the height contrast. Uncontrolled height-correlated biases remain degenerate with anomalous redshift. This is a theory baseline, not a new experiment, an empirical limit, or a derivation of spacetime.

1 Geometry acts on clock rate; radiation acts on local spectrum

Use the test-clock approximation: tidal effects across the ion and heater-induced changes in the background are negligible or independently bounded. Assume a static background with

$$ds^2 = -N(\mathbf{x})^2 c^2 dt^2 + h_{ab} dx^a dx^b. \quad (1)$$

For a clock held at rest, $d\tau = N dt$. A local transition with proper frequency ν_i^{loc} accumulates phase $d\varphi_i = 2\pi\nu_i^{\text{loc}} d\tau$. Stationary phase transfer to a reference at R therefore gives

$$\frac{\nu_{i,\text{received}}^S}{\nu_3^R} = \frac{\nu_i^0}{\nu_3^0} \frac{N_S}{N_R} \frac{1 + b_i^S}{1 + b_3^R} \exp(A_i^S - A_3^R + \ell_i). \quad (2)$$

Here A denotes other local log-frequency shifts and ℓ_i nonideal transfer corrections. They are not silently zero in data. G-R4's local dipole response is unchanged:

$$b_i^X = -\frac{1}{2h\epsilon_0\nu_i^0} \mathcal{P} \int_0^\infty \Delta\alpha_i(\omega) u_\omega^X d\omega, \quad B_i^X = \ln(1 + b_i^X). \quad (3)$$

For the retained ideal static approximation, $b_i(T) = \beta_i a_{\text{rad}} T^4$ and $\beta_i = -\Delta\alpha_i/(2h\epsilon_0\nu_i^0)$. All temperatures and spectra are local proper quantities.

Normalize Eq. (2) by ν_i^0/ν_3^0 and take its log:

$$q_i = G + B_i^S - B_3^R + A_i^S - A_3^R + \ell_i + n_i, \quad G = \ln(N_S/N_R). \quad (4)$$

The universal gravitational factor is now specified by geometry, not a fitted replacement for G-R5's unidentified common factor. To first weak-field order, $\Delta G = \Delta\Phi/c^2$ for stationary nonrotating coordinates. For moving clocks an additional $-\Delta(v^2)/(2c^2)$ appears. This is a restricted redshift/local-position-invariance baseline, not a test of every classical or quantum EEP component [1].

2 Four settings, three contrasts

Use the same target ion and enclosure, with E2/E3 interleaved at each settled plateau. Keep the independent E3 reference at its fixed site. Let $a = 0, 1$ label local temperature and $b = 0, 1$ label potential. The column order is (00, 10, 01, 11). After evaluating reference, transfer and non-BBR shifts, the ideal model is

$$q_{i,ab} = q_{i,00} + a\theta_i + b\gamma, \quad \theta_i = B_i(T_1) - B_i(T_0), \quad \gamma = \ln \frac{N_1}{N_0}. \quad (5)$$

This holds for any fixed local $B_i(T)$, not only its static T^4 approximation, provided the same local spectrum and apparatus conditions are reproduced at both potential settings. Define

$$\mathcal{T}_i = \frac{1}{2}(q_{i,10} - q_{i,00} + q_{i,11} - q_{i,01}) = \theta_i, \quad (6)$$

$$\mathcal{G}_i = \frac{1}{2}(q_{i,01} - q_{i,00} + q_{i,11} - q_{i,10}) = \gamma, \quad (7)$$

$$\mathcal{I}_i = q_{i,11} - q_{i,10} - q_{i,01} + q_{i,00} = 0. \quad (8)$$

Thus the null predictions are

$$\mathcal{G}_2 = \mathcal{G}_3 = \gamma, \quad \mathcal{G}_3 - \mathcal{G}_2 = 0, \quad \mathcal{I}_2 = \mathcal{I}_3 = 0. \quad (9)$$

The internal E3/E2 ratio cancels gravity: it cannot by itself measure the common redshift. The outside reference is essential.

One illustrative calculation, not observations. Retain the pinned G-R4 polarizabilities, $T_0 = 300$ K and $T_1 = 310$ K. Take $\Delta h = 1$ m and conventional $g_* = 9.80665$ m s⁻² merely as a scale illustration, not measured laboratory geodesy. Then $\gamma \simeq g_* \Delta h / c^2 = 1.0911 \cdot 10^{-16}$:

Setting	$(q_2 - q_{2,00})/10^{-16}$	$(q_3 - q_{3,00})/10^{-16}$
T_0, Φ_0	0	0
T_1, Φ_0	-0.7337	-0.1012
T_0, Φ_1	1.0911	1.0911
T_1, Φ_1	0.3574	0.9899

The thermal E3/E2 contrast is $6.325 \cdot 10^{-17}$; the gravitational E3/E2 contrast is zero. All entries are model predictions.

Why use logarithms? For the baseline-normalized raw ratio $r_{ab} = \exp(a\theta_i + b\gamma)$, the raw cross contrast is

$$r_{11} - r_{10} - r_{01} + r_{00} = (e^{\theta_i} - 1)(e^\gamma - 1), \quad (10)$$

not zero. This ordinary multiplicative product is not new physics. Logarithms make the separability null exact within the stated model.

3 What cancels, what does not, and identifiability

An arbitrary but height-independent $B_i(T_a)$ cancels in $\mathcal{G}_i$. Consequently the old polarizability uncertainty need not impose its full G-R5 thermal-composite floor on the matched-bath height test. This is not permission to neglect radiation matching. If the higher site instead has $T_a + \delta T_a$, then

$$\delta \mathcal{G}_i^{\text{BBR}} \simeq \frac{1}{2} \sum_{a=0}^1 B'_i(T_a) \delta T_a, \quad B'_i(T) = \frac{4b_i(T)}{T[1 + b_i(T)]} \quad (11)$$

in the retained static model. The exact formula is one half the sum of $B_i(T_a + \delta T_a) - B_i(T_a)$. An illustrative +0.1 K mismatch at both upper-site setpoints leaks $-7.345 \cdot 10^{-19}$ for E2 and $-1.013 \cdot 10^{-19}$ for E3. Dynamic, anisotropic and geometry changes require their own responses; matching a wall thermometer alone does not establish matching spectra.

G-R5's composite weights

$$w_2 = \frac{\beta_3}{\beta_3 - \beta_2}, \quad w_3 = -\frac{\beta_2}{\beta_3 - \beta_2}, \quad (w_2, w_3) \simeq (-0.16004, 1.16004) \quad (12)$$

remain a leading thermal-rejection diagnostic. They obey $w_2 + w_3 = 1$ and $w_2\beta_2 + w_3\beta_3 = 0$, so the common height contrast is retained. Their use is not necessary for the exact matched-bath cancellation above.

For interpretation only, parametrize a possible anomalous height response by α_i :

$$\mathcal{G}_i = (1 + \alpha_i)\gamma + c_h + \varepsilon_i. \quad (13)$$

The baseline is $\alpha_i = 0$, not a fit. A shared height-correlated reference/link bias c_h then gives

$$D_h = \mathcal{G}_3 - \mathcal{G}_2 = (\alpha_3 - \alpha_2)\gamma + \varepsilon_3 - \varepsilon_2, \quad (14)$$

$$C_h - \gamma = (w_2\alpha_2 + w_3\alpha_3)\gamma + c_h + w_2\varepsilon_2 + w_3\varepsilon_3. \quad (15)$$

For $\gamma \neq 0$, shifting both α_i by λ and c_h by $-\lambda\gamma$ leaves the observations unchanged. A common anomaly is therefore unidentifiable without independent bias bounds. Differential biases can also mimic D_h ; a nonzero interaction contrast only falsifies the controlled separability model, not EEP by itself.

Order the eight channel means as E2's four cells followed by E3's. For any contrast vector a , its variance is $a^T V a$. A shared, cell-varying reference contribution has covariance $V_R = \mathbf{1}_2 \mathbf{1}_2^T \otimes K_R$. It cancels for $a_D = (-1, 1) \otimes (-1, -1, 1, 1)/2$, but not generally for $a_C = w \otimes (-1, -1, 1, 1)/2$. A constant reference offset cancels from both. Matched effective sampling windows are required.

4 Independent controls and a bounded empirical handoff

Potential. Measure $\Delta\Phi$ independently at the ion positions using leveling/gravimetry/geodesy with covariance. A portable-clock campaign with independent geodetic comparison [2] reports the magnitude $3915.88(0.30) \text{ m}^2 \text{ s}^{-2}$. This gives a source-backed baseline $|\Delta G| \simeq 4.3570 \cdot 10^{-14}$ with propagated potential uncertainty $3.34 \cdot 10^{-18}$. We use the geodetic value, not the clock-inferred potential. The published clocks were Sr: these are not our E2/E3 four-setting data or a calibration of our apparatus. For this upward-positive convention, $\Delta\Phi = \int_{h_0}^{h_1} g(h)dh$ locally. If a terrestrial geopotential already includes centrifugal effects, do not add the corresponding rotational kinetic term again. Specify sign and transfer convention; stationary-plateau frequency comparisons are not transported-clock accumulated-phase measurements. A 10^{-18} absolute redshift error corresponds to $\sigma_{\Delta\Phi} \simeq 0.0899 \text{ m}^2 \text{ s}^{-2}$, or about 0.92 cm at g_* , for this contribution alone. No such accuracy is claimed. In a residual $r = \mathcal{G} - \gamma_{\text{cal}}$, propagate $\text{Var } r = \text{Var } \mathcal{G} + \text{Var } \gamma_{\text{cal}} - 2 \text{Cov}(\mathcal{G}, \gamma_{\text{cal}})$.

Temperature and reference. Use independent local radiometry, with sensor calibration, gradients, emissivity/view factors and reference monitoring at every plateau. Do not infer the bath from the tested frequency law, or infer potential from the same clock ratios and then call the prediction verified. Radiometry/correction covariance enters V through its response Jacobian. Actual measured settings, not ideal labels, enter the model.

Equilibrium is a different boundary condition. Independently thermostatted local setpoints need not form one global passive equilibrium bath. In a static globally equilibrated radiation field, conservation of Killing energy gives $E_\infty = N E_{\text{loc}}$; the Bose factor written locally therefore has

$$e^{E_\infty/(k_B T_\infty)} = e^{E_{\text{loc}}/(k_B T_{\text{loc}})}, \quad T_{\text{loc}} N = T_\infty. \quad (16)$$

This is the Tolman–Ehrenfest relation [3, 4]. Do not impose it simultaneously with freely chosen local temperatures. At 300 K across the illustrative 1 m separation it predicts only $\Delta T \simeq -300\gamma = -3.27 \cdot 10^{-14} \text{ K}$. This observation neither changes the imposed 10 K control nor requires resolving that tiny equilibrium gradient.

Acquisition and stopping rule. One optional symmetric order is (00, 10, 11, 01, 01, 11, 10, 00) at centers $(-7, -5, -3, -1, 1, 3, 5, 7)\tau$. Pair averaging cancels ideal linear drift in all three contrasts, but quadratic drift survives: its $(\mathcal{T}, \mathcal{G}, \mathcal{I})$ moments are $(-8, -32, 32)\tau^2$. State-locked bias also survives. Actual settling/windows, return comparisons, reversed orders, transport-systematic monitoring and repeated blocks remain necessary. Four means do not determine statistical significance. The input card records the missing acquisition fields. No paired data, new experiment or lab contact is supplied; no time law or Route-F gate is promoted. The baseline is complete within its assumptions.

References

- [1] M. Zych and C. Brukner, Nature Physics 14, 1027 (2018).
- [2] J. Grotti et al., Physical Review Applied 21, L061001 (2024).
- [3] R. C. Tolman and P. Ehrenfest, Physical Review 36, 1791 (1930).
- [4] N. Karolinski and V. Faraoni, European Physical Journal C 84, 1248 (2024), section 1.